

# Design Prompts that Improve Performance



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# Overview



**Factors That Impact Latency**  
**Improve Prompt Design**





# **Factors That Impact Latency**



# Latency

Prompt



Input



Latency

Generated Text

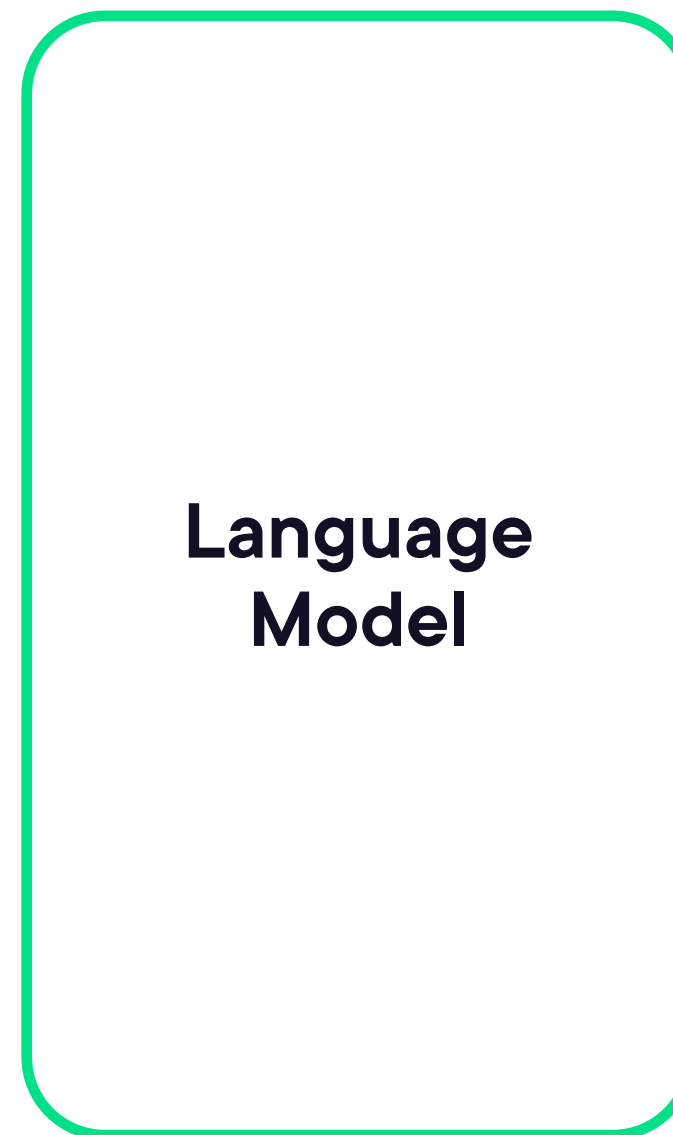


Output

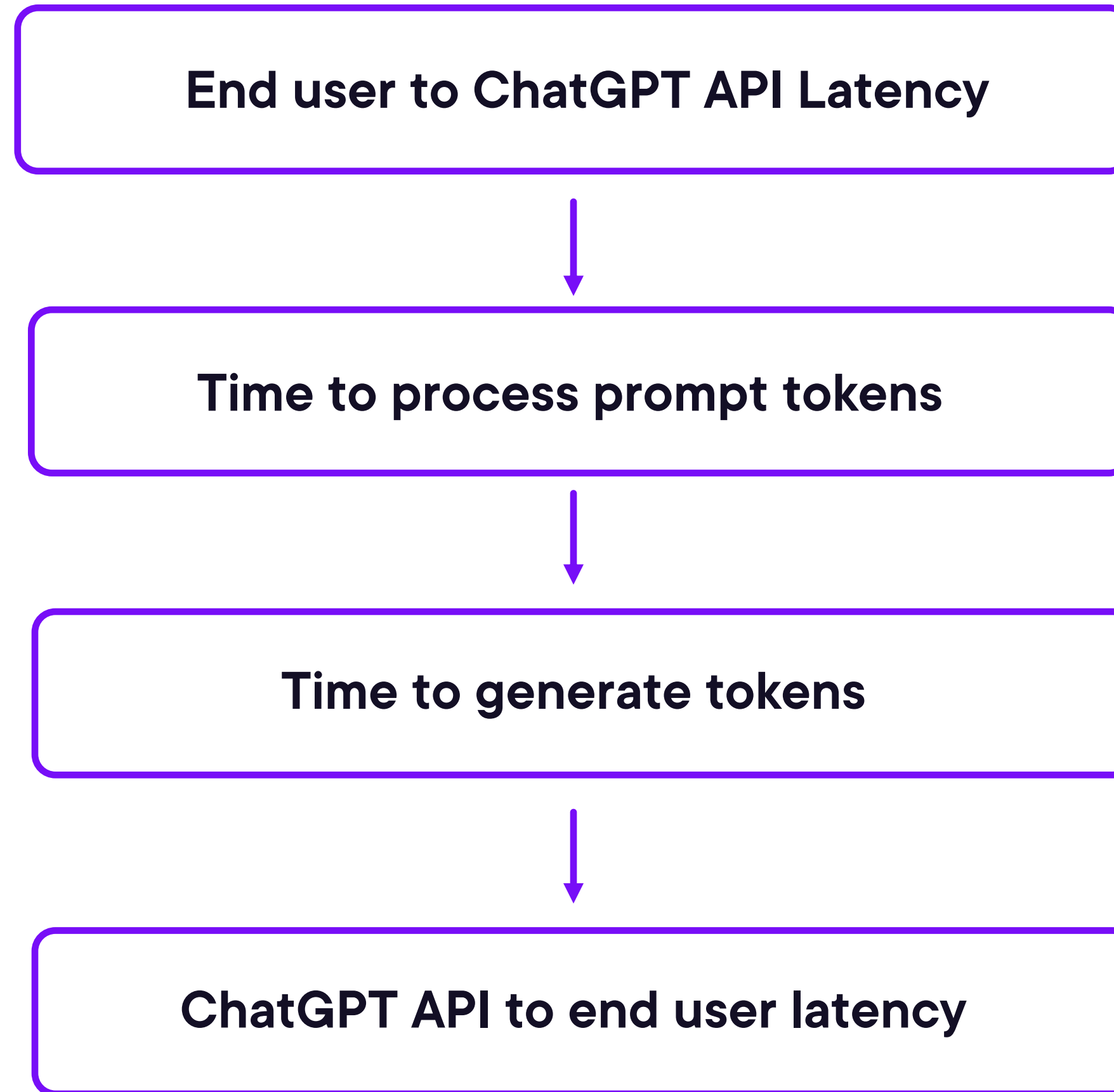


Latency

Language  
Model



# Flow of Completion Request



# Tokens

A common sequence of characters found in text.

Tokenization = Tokenization

The = The



# Tokenization Prompt

Translate the following sentence from English to Italian: I love  
learning new things!

16 tokens



# Factors Impacting Latency

**Model**

**Number of completion tokens**





## Models

OpenAI offers different models with varying levels of complexity and generality.





## Number of Completion Tokens

Requesting a large amount of generated tokens completions can lead to increased latencies.



**Up Next:**

# **Improve Prompt Design**

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# Improve Prompt Design



# Model Selection

## Newer GPT Models

**More capable**

**Produces more complex and diverse output**

**Increased latency**

**Slower and more expensive**

**VS**

## Older GPT Models

**Less capable**

**Produces less accurate or relevant output**

**Lower latency**

**Faster and cheaper**



# Reduce Generated Tokens

**Stop sequences**

**max\_token  
attribute**

**n and stream  
attributes**



# Stop Sequences

Recommend three books in the mystery genre.



stop sequence



```
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=messages,  
    max_tokens=15  
)
```

## **max\_tokens Parameter**

**This reduces the number of tokens returned in the response.**



```
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=messages,  
    n=3  
)
```

## n Parameter

The number of chat completion choices to generate.

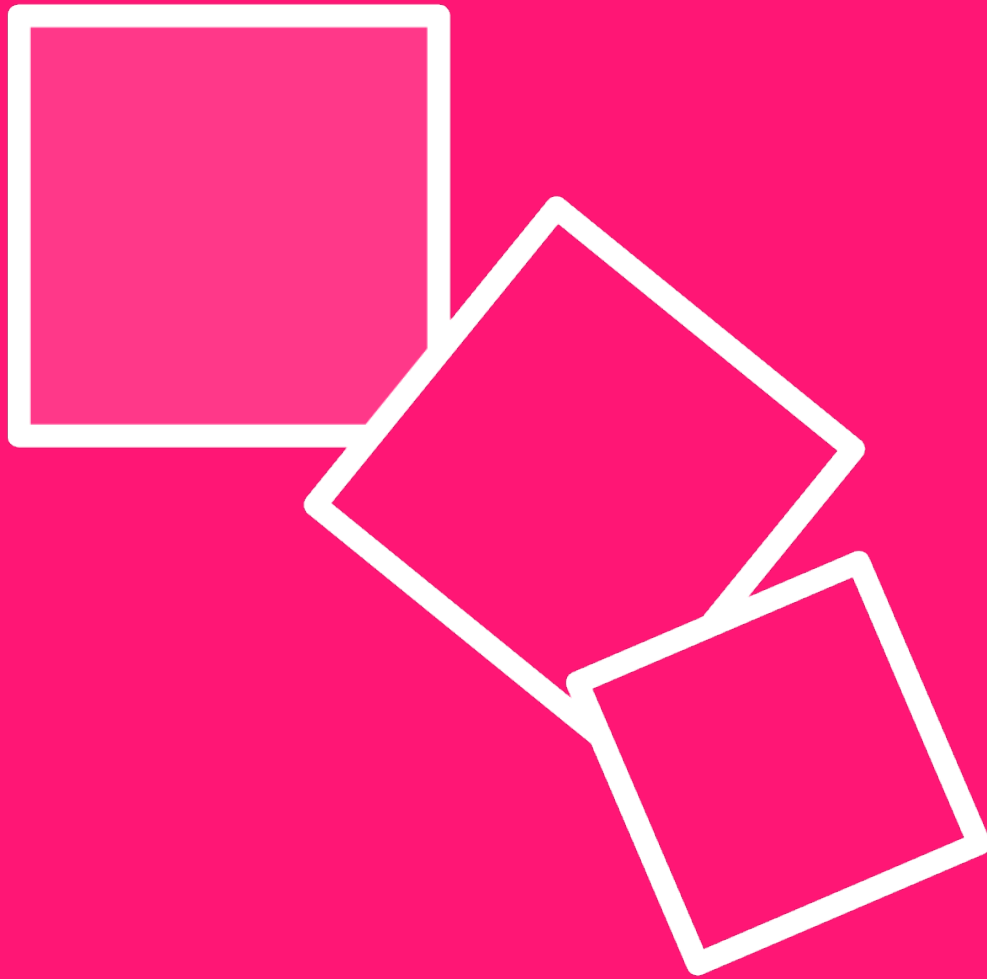


```
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=messages,  
    stream=True  
)
```

## stream Parameter

Returns tokens as soon as they are available.

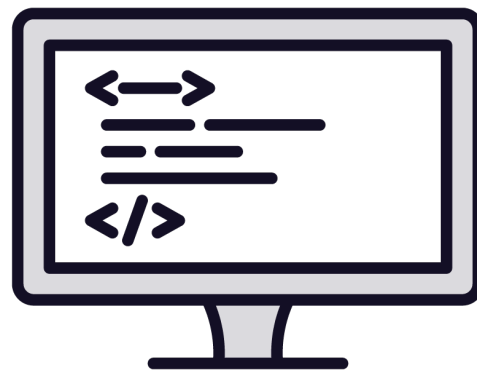




# Track and Version Prompts

Discover better prompts over time through tracking and versioning.





## Hands-on Challenge

**Update one of the prompts from the Summarizing and Expanding Text lesson to reduce the number of tokens returned.**



**Up Next:**

# **Course Conclusion**

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# Module Conclusion



**Factors That Impact Latency**  
**Improve Prompt Design**



# Course Conclusion



**Harness capabilities of language models**

**Advanced prompting techniques**

- Chain-of-thought
- Knowledge augmentation
- Prompt tuning

**Prompt optimization**

- Summarizing and expanding text
- Text-to-image generation
- Inferring sentiment
- Text transformation

**Reduce latency and improve performance**

